Deterministic Document Layout at Zero Dependencies

An evaluation of the OmniPDF convergence paginator — measuring how far a strict block model, real font metrics, and a fixpoint footnote algorithm take a dependency-free engine.

1. Introduction

Programmatic PDF generation splits into two camps: writers that place glyphs at absolute coordinates, and layout engines that flow structured content across pages. The former are simple and predictable; the latter are where documents actually live. This report describes a paginator that treats page composition as a convergence problem rather than a single-pass measure-and-place sweep.

The central observation is that footnotes, table headers, and keep rules are **coupled constraints**: adding a footnote can push its own reference across a page boundary, at which point the note must move too[1]. A monotone iteration with an explicit non-convergence error resolves it in bounded time.

2. Results

The table below spans multiple pages; its header repeats automatically on every continuation page, and no row ever splits across a boundary.

Engine property	Single-pass	Convergence	Delta
Property 1 — measured across the full corpus of documents	80%	97%	+0pt
Property 2 — measured across the full corpus of documents	87%	99%	+3pt
Property 3 — measured across the full corpus of documents	94%	98%	+6pt
Property 4 — measured across the full corpus of documents	82%	97%	+9pt
Property 5 — measured across the full corpus of documents	89%	99%	+12pt
Property 6 — measured across the full corpus of documents	96%	98%	+15pt
Property 7 — measured across the full corpus of documents	84%	97%	+1pt
Property 8 — measured across the full corpus of documents	91%	99%	+4pt

[1] This is the classic fixpoint of document layout; TeX solves it with multiple passes, Word with a similar iterative scheme.

Engine property	Single-pass	Convergence	Delta
Property 9 — measured across the full corpus of documents	98%	98%	+7pt
Property 10 — measured across the full corpus of documents	86%	97%	+10pt
Property 11 — measured across the full corpus of documents	93%	99%	+13pt
Property 12 — measured across the full corpus of documents	81%	98%	+16pt
Property 13 — measured across the full corpus of documents	88%	97%	+2pt
Property 14 — measured across the full corpus of documents	95%	99%	+5pt
Property 15 — measured across the full corpus of documents	83%	98%	+8pt
Property 16 — measured across the full corpus of documents	90%	97%	+11pt
Property 17 — measured across the full corpus of documents	97%	99%	+14pt
Property 18 — measured across the full corpus of documents	85%	98%	+0pt
Property 19 — measured across the full corpus of documents	92%	97%	+3pt
Property 20 — measured across the full corpus of documents	80%	99%	+6pt
Property 21 — measured across the full corpus of documents	87%	98%	+9pt
Property 22 — measured across the full corpus of documents	94%	97%	+12pt
Property 23 — measured across the full corpus of documents	82%	99%	+15pt
Property 24 — measured across the full corpus of documents	89%	98%	+1pt
Property 25 — measured across the full corpus of documents	96%	97%	+4pt
Property 26 — measured across the full corpus of documents	84%	99%	+7pt
Property 27 — measured across the full corpus of documents	91%	98%	+10pt

Engine property	Single-pass	Convergence	Delta
Property 28 — measured across the full corpus of documents	98%	97%	+13pt
Property 29 — measured across the full corpus of documents	86%	99%	+16pt
Property 30 — measured across the full corpus of documents	93%	98%	+2pt
Property 31 — measured across the full corpus of documents	81%	97%	+5pt
Property 32 — measured across the full corpus of documents	88%	99%	+8pt
Property 33 — measured across the full corpus of documents	95%	98%	+11pt
Property 34 — measured across the full corpus of documents	83%	97%	+14pt
Property 35 — measured across the full corpus of documents	90%	99%	+0pt
Property 36 — measured across the full corpus of documents	97%	98%	+3pt
Property 37 — measured across the full corpus of documents	85%	97%	+6pt
Property 38 — measured across the full corpus of documents	92%	99%	+9pt
Property 39 — measured across the full corpus of documents	80%	98%	+12pt
Property 40 — measured across the full corpus of documents	87%	97%	+15pt
Property 41 — measured across the full corpus of documents	94%	99%	+1pt
Property 42 — measured across the full corpus of documents	82%	98%	+4pt

3. Unicode through embedded fonts

Base-14 fonts cover Western Europe. Everything else travels through embedded, **subsetted TrueType**: Русский текст, Ελληνικά, and English share one measurer, one paginator, one renderer.

Reproduction: `npx tsx examples/report.ts`. The bytes you get are the bytes everyone gets[2].

[2] Determinism is asserted in CI: the same document model must produce byte-identical output on every platform.